

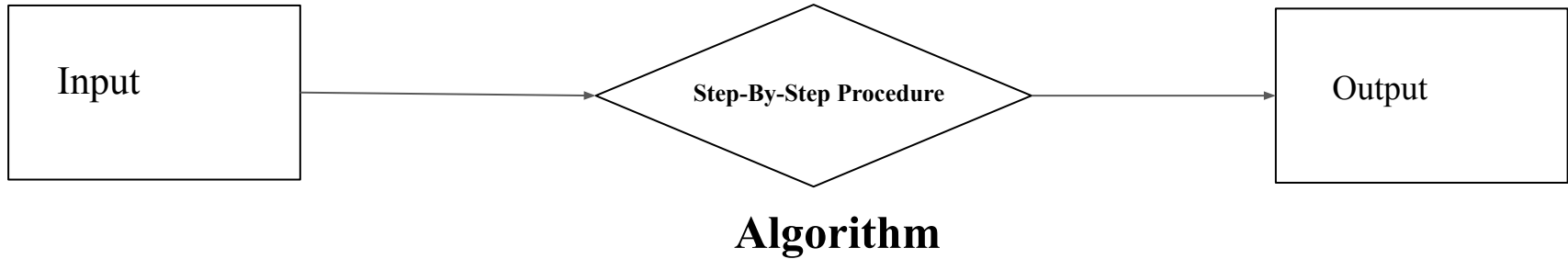
Algorithm Properties & Complexity Analysis

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Algorithm

Q: What is Algorithm?

Answer: An algorithm is a step-by-step procedure or a set of instructions designed to perform a specific task or solve a particular problem.



Necessity Algorithm

Q: What is the need for algorithms?

Answer: Algorithms are essential for several key reasons in computer science.

1. **Efficiency:** Algorithms help in solving problems in an efficient manner.
2. **Problem-Solving:** Algorithms are a framework for solving problems systematically.
3. **Scalability:** Well-designed algorithms allow systems to handle increasing loads of data.
4. **Standardization:** Algorithms provide a standardized way to solve specific types of problems.
5. **Optimization of Costs:** Algorithms help achieve these goals through optimization techniques.
6. **Complex Problem Solving:** Some problems, especially in fields like cryptography, optimization, and artificial intelligence, are highly complex and require sophisticated algorithms to be solved.

Algorithm Criteria

Q: What are the criteria for an algorithm?

Answer: The main criteria are:

- 1.**Input:** Zero or more quantities are externally supplied.
- 2.**Output:** At least one quantity is produced.
- 3.**Definiteness:** Each instruction is clear.
- 4.**Finiteness:** The algorithm terminates after a finite number of steps.
- 5.**Effectiveness:** Every instruction must be basic and feasible.

Types of Algorithms

- 1.Brute Force Algorithm
- 2.Recursive Algorithm
3. Backtracking Algorithm
- 4.Searching Algorithm
- 5.Sorting Algorithm
- 6.Hashing Algorithm
- 7.Greedy Algorithm
- 8.Dynamic Programming Algorithm

Advantages of Algorithms

1. Easy to Understand
2. Step-by-Step Representation
3. Simplifies Complex Problems
4. Language Independence
5. Easy to Debug and Test
6. Improves Efficiency & Accuracy
7. Optimization

Complexity Analysis

Complexity analysis evaluates the efficiency of an algorithm based on its resource consumption.

Why it needs:

- 1.Ensures efficient resource usage.
- 2.Helps in comparing algorithms.
- 3.Critical for scalability with large inputs.

Types of Complexity

1. Time Complexity: Measures how the algorithm's running time grows as the input size increases.
2. Space Complexity: Measures how much memory the algorithm needs as the input size increases.

Asymptotic Notation

Big-O Notation (O): Worst-case scenario.

Omega Notation (Ω): Best-case scenario.

Theta Notation (Θ): Average-case or tight bound.

Best, Worst, and Average Cases

Best-case: Minimum time for most favorable input.

Worst-case: Maximum time for the least favorable input.

Average-case: Expected time for random inputs.

Example: Time & Complexity of Merge Sort

Best, Worst, and Average Case: $O(n \log n)$

Space Complexity: $O(n)$ (due to temporary arrays)